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Development of a Hydrogen Microgas Turbine Combustor: NO_x Emissions and Secondary Air Injection

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On the way to defossilization, green hydrogen is a promising way to substitute natural gas (NG) and oil in the gas turbine industry. In the scope of the H2mGT project, a microgas turbine (mGT) burner with 100% hydrogen firing is developed and validated. The project is funded by the German BMWK, and it is a collaboration between Technische Universität Berlin (TUB) and the manufacturer Euro-K GmbH. It consists of three phases: (1) atmospheric pressure tests with a fused silica combustion chamber; (2) atmospheric pressure tests with counterflow-cooled steel flame tube and secondary air injection; (3) validation of the burner in the mGT at elevated pressure levels. The current study will present the results of phase 2. The hydrogen burner used in the project is based on a swirl-stabilized burner of TUB and was scaled to 36 kW thermal power at atmospheric conditions. The burner design features a variable swirl intensity, additional axial momentum of air in the mixing tube, a movable central fuel lance, and pilot nozzles at the front plate. Furthermore, the steel flame tube is exchangeable, which allows the evaluation of different dilution hole patterns and, thus, the variation of the ratio of primary and secondary air. The study presents temperature, pressure, and emission measurements. It is found that the flame can be operated over a large range of equivalence ratios and preheating temperatures up to 500 °C. As expected, the NO_x emissions are mainly influenced by the local equivalence ratio, which can be controlled by the fuel mass flow or the dilution hole pattern in the flame tube. Furthermore, the results show a decrease of NO_x when the power density is increased at constant equivalence ratios, and a rise of NO_x during the fuel transition from natural gas to hydrogen. The results indicate certain differences to the findings of phase 1.
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Keywords: microgas turbine, combustion, hydrogen, emissions

1 Introduction

The current developments in the global climate crisis make a rapid shift away from fossil fuels inevitable. Nevertheless, combustion engines—especially gas turbines—will continue to be an essential part of our energy supply thanks to their flexibility and reliability. A CO_2-free alternative to fossil fuels is "green hydrogen" produced from renewable energies. The joint H2mGT project of Technische Universität Berlin (TUB) and the microgas turbine manufacturer Euro-K GmbH aims to develop a hydrogen (H_2) combustor for a microgas turbine (mGT).

In the industry, almost all major manufacturers are developing various combustion systems for syngas and H_2-enriched natural gas. In general, increased NO_x emissions and problems with flame stability are reported, so the H_2 capability is still limited at the commercial level. A recent overview about the developments in the industry is given by the ETN Global's Hydrogen Gas Turbines report [1] and Cecere et al. [2].

The mGT is a perfect platform to investigate H_2 combustion on a machine level since the necessary investment, the operating pressure, and the exhaust gas temperatures are much lower than for heavy-duty gas turbines. In addition, mGTs usually offer good accessibility to the combustion chamber. They provide an alternative to gas piston engines in the small and medium combined heat and power segment. They are very flexible and emit significantly less pollutants than piston engines [3]. Modern mGT combustor systems achieve NO_x emissions of 15 ppm, while current lean-burn engines in the medium power class with oxidation catalyst emit NO_x values of 125–250 ppm. The German Aerospace Center [3] predicts a large potential for efficiency improvements and a rising share on the combined heat and power market for mGTs. A review of numerical studies with respect to different mGT-based energy systems is given by Reale and Sannino [4]. On the experimental side, mGT combustors have been investigated for a

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variety of unconventional fuels [5,6], including H_2. In a small mGT (1–3 kW_{el}), du Toit et al. [7] measured emissions of $H_2/CH_4/CO_2$ fuel blends. Adding 15%$_{vol}$ H_2 to CH_4 showed no significant effect on the NO_x emissions. The study also reports the influence of the fuel switch on the mGT operation and auxiliaries. On a larger mGT (Capstone C30, 30 kW_{el}), de Santoli et al. [8] tested H_2 addition in natural gas up to 10%$_{vol}$. In this study, the H_2 was used to compensate the engine derating due to increased inlet temperature in the summertime. A slight increase of the NO_x is reported for the H_2 enrichment. The UCI Combustion Laboratory retrofitted a Capstone C60 mGT [9,10] for the operation with H_2. Despite adjustments to the fuel injection, the tests showed significantly increased NO_x emissions. Devriese et al. [11] also found high NO_x emissions for an mGT burner with H_2 firing in numerical studies. Recently, a T100 microgas turbine was successfully tested with H_2 enrichment and even pure hydrogen by Banihabib and Assadi [12] in Stavanger. The original combustor was replaced by a fuel-flexible jet-stabilized burner, which was developed and lab-tested by the German Aerospace Center in Germany [13]. Stable operation in the mGT was achieved up to 100% H_2. An incline in NO_x was observed resulting in 62 ppm (15% O_2) at base load [14].

In summary, there are many challenges in the combustion of H_2. On the one hand, H_2 has a lower volumetric energy density (Table 1), which requires a higher fuel flow. On the other hand, H_2 has a high gravimetric energy density and a significantly higher reactivity compared to natural gas, which might increase the flame temperature and speed. The former can have a negative impact on NO_x emissions, while the latter imposes a risk of flashback for premixed flames. These challenges are to be tackled in the present project with a swirl-stabilized dry low NO_x burner with axial air injection. The TUB has investigated a similar burner in numerous projects with H_2 and enriched natural gas and was able to achieve very good results regarding the prevention of flame flashbacks and the minimization of NO_x emissions [15–23]. The burner design of the current project is presented in detail in Sec. 2.

The H2mGT project is divided into three phases. First, the base combustor is scaled and redesigned according to the requirements of the mGT. It is then tested in an atmospheric combustion test rig with a fused silica combustion chamber using OH* flame recording and emission analysis. In phase 2, the transparent chamber is replaced by a steel flame tube with counterflow cooling and secondary air injection. Using this setup, the burner is optimized in the test rig regarding emissions and stable operation. Once this phase is complete, the burner is adapted for installation in the MTG35 microgas turbine at the Euro-K test facility. The mGT has 130 kW thermal power (by fuels lower heating value (LHV)), 3.6 pressure ratio, up to 28% electrical efficiency and is recuperated to 500 °C combustor inlet temperature. In phase 3, the pressure is increased and the new burner is validated under nominal operating conditions. A previous paper presented the results of the project's phase 1 [24], which is followed by the current study, in which the results of phase 2 are analyzed.

2 Burner and Test Rig Design

The burner designed for this project is based on a generic swirl-stabilized burner, which was already extensively studied at TUB [15–23]. It was scaled to the thermal power of the mGT by maintaining similar volumetric flows in the burner. Geometrical parameters in the burner design are supposed to allow for an optimization of the operation range and NO_x emissions.

Table 1 Fuel properties

Property	H_2	Natural gas	CH_4 ref	Unit
LHV mass	119.97	47.07	50.01	MJ/kg
LHV volume	10.78	37.51	35.88	MJ/m^3
Molar mass	2.016	—	16.04	g/mol
Wobbe index	40.90	47.81	48.17	MJ/m^3
α	0	—	1	
β	0.5	—	1	

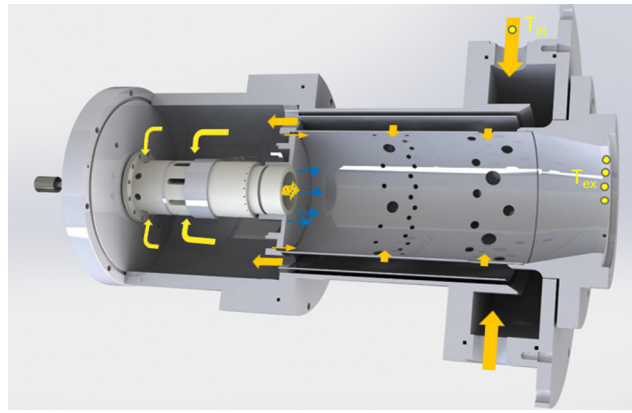

Fig. 1 Schematic of the phase 2 burner with secondary air injection

Figures 1 and 2 show a schematic of the burner in phase 2. The preheated air is fed to a preplenum via four inlets downstream of the combustion chamber. The air then streams as counterflow along the outer flame tube wall. The stainless steel flame tube replaces the fused silica combustion chamber from phase 1 and allows secondary air injection through a pattern of dilution holes. Consequently, the air mass flow is divided into primary air, which takes part in the combustion, and secondary air, which dilutes the high-temperature exhaust gas. The primary air flows through the mixing tube ($D_{mt} = 31$ mm) into the combustion chamber ($D_{cc} = 100$ mm, $L_{cc} = 180$ mm). It is fed from the plenum in three ways: a tangential flow (via the swirler), an axial flow in the center (left orange arrows in Fig. 1) and small wall jets further downstream to lean out the boundary layer. In the case of the secondary air, the largest part is directly flowing into the combustion chamber through the dilution holes in the flame tube. The flow turns 180 deg at the holes and penetrates as jets in the combustion chamber flow as shown in Fig. 3. A smaller part is injected parallel to the liner to support the cooling of the flame tube wall. The actual distribution of the air will be discussed in Sec. 4.

The fuel can be injected through three different injection points (numbers in Fig. 2):

(1) Premixed: 16 nozzles ($D_{P1} = 1.4$ mm) at the bottom of the mixing tube, which is regarded as technically premixed,
(2) Central: an axial nozzle ($D_{P2} = 3$ mm) fuel lance ($D_{cb} = 6$ mm) in the mixing tube as central injector,
(3) Pilot: eight pilot nozzles ($D_{P3} = 0.8$ mm) at the front plate around the mixing tube exit.

The latter are supplied by a cavity behind the burner plate. The pilot is intended for ignition in the mGT and as emergency backup for flashback recovery. The idea of the central injection via the fuel

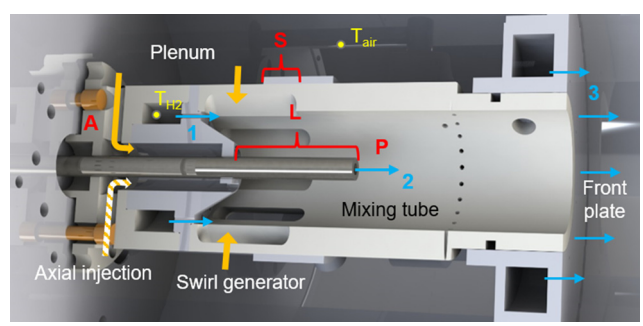

Fig. 2 Detailed view of the mixing tube and the geometric parameters

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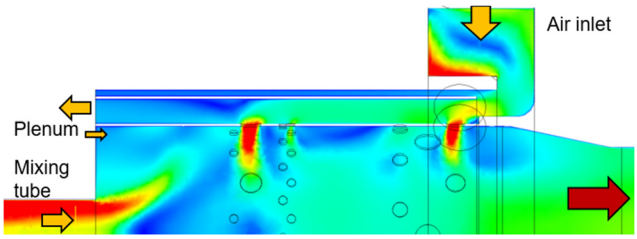

Fig. 3 Qualitative flow field of the absolute velocity in the combustion chamber from a RANS simulation

Table 2 Fuel injector prepressure

Injector	H_2 mass flow in kg/h	Pressure in barA
Premixed	1.36	1.785
Central	1.36	1.635
Pilot	1.36	2.566

lance is that the H_2 fraction profile in the mixing tube can be altered. The three fuel injectors are supplied from a H_2 bundle via two mass flow controllers, one for premixed and one for either central or pilot injection (which can be switched during operation by a three-way ball valve). The injector prepressure is measured by the mass flow controllers and displayed in Table 2.

The burner features four parameters to alter the flow field and fuel distribution. These parameters are depicted in Fig. 1 and can be adjusted manually from outside without disassembling the burner. Hence, the design points in an optimization can be set during the operation of the burner, which allows for the evaluation of much more configurations. The parameters are (letters in Fig. 2):

- S—swirler blockage in mm,
- A—axial injection restriction in # of blocked channels,
- L—length of the center body in mm, and
- P—share of either central or pilot fuel mass flow %.

The swirler consists of eight tangential slots ($L_s = 27$ mm), which can be blocked partially (S). The axial air injection flows from a small subplenum through an annulus around the center body into the mixing tube. The subplenum is fed by eight radial channels from the main plenum. These channels can be separately blocked (A). The fuel lance is movable in the axial direction (L) and can be positioned between 0 mm and 90 mm, the latter corresponding to the end of the mixing tube. The variation of this position has two effects. On the one hand, it changes the flow field in the mixing tube and the combustion chamber because the lance body creates a wake in the flow. On the other hand, the point of the central fuel injection can be moved downstream leading to a stronger impact on the fuel distribution in the combustion chamber. The parameter P is the share of the central or the pilot fuel injection. Hence, $P = 0\%$ corresponds to the mode, when only the premixed nozzles are activated, while $P = 100\%$ means that all the fuel flows through either the central or the pilot injection. Please note that central and pilot injections are never used at the same time.

The flame is ignited by a spark plug in the combustion chamber 56 mm downstream of the front plate. A nozzle at the end reduces the diameter from the 100 mm combustion chamber to a 80 mm fused silica tube, which is followed by a water-cooled exhaust tube. Four type K thermocouples at the end of the nozzle measure the exhaust temperature at different radii. A small portion of the hot gas is sampled downstream in the exhaust tube by a water-cooled multihole probe. A gas analysis unit measures the NO_x content (± 1 ppm) for emission evaluation.

3 Measurement Settings

The mGT has a thermal power of 130 kW and a nominal air mass flow of 1280 kg/h. However, these values are scaled down by

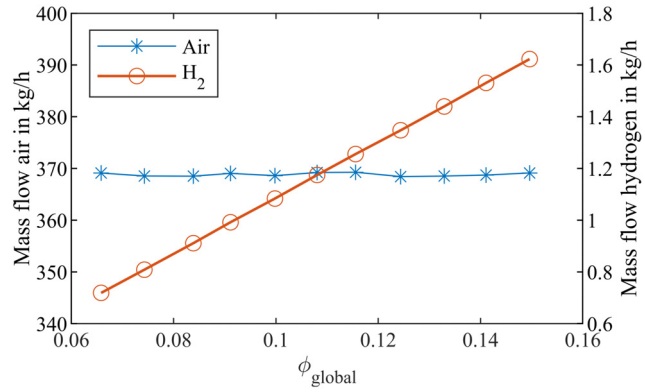

Fig. 4 Air and H_2 mass flows of a measurement series

pressure ratio (constant Ma number) since the tests are conducted at atmospheric pressure. The air mass flow is fixed to $\dot{m}_{air} = 368$ kg/h for most of the experiments. In the mGT, the total air mass flow at baseload conditions is fixed by the output of the compressor, but the air is split into primary and secondary air. Only the primary air flows through the combustion zone and takes part in the reaction. The local equivalence ratio of the flame can be varied by changing the splitting ratio (e.g., changing the hole pattern in the flame tube). In contrast, the global equivalence ratio remains the same for all flame tubes, but can be varied via the hydrogen mass flow, as shown in Fig. 4. The bounds are set to $\phi_{global} = 0.06$ and 0.15 or $\dot{m}_{H_2} = 0.7$ and 1.6 kg/h, respectively. The base load conditions of the mGT correspond to a ϕ_{global} of 0.10–0.12. The air is preheated to 500 °C in the plenum to match the conditions of the recuperated mGT. The fuel is supplied at room temperature but the fuel heats up while flowing through the hot metal body of the burner. The actual fuel temperature of the premixed nozzles is measured by a thermocouple in the fuel plenum (Fig. 2) and is shown exemplarily in Fig. 5. The operating conditions are summed up in Table 3.

The mass flows and temperatures of a typical measurement series are shown in Figs. 4 and 5. Please note that the air is preheated by two separate electrical heaters, of which one reaches $T_{in} = 500$ °C and the other only 450–460 °C. Therefore, T_{air} refers to the unified air temperature in the plenum, which can be even higher than 500 °C for certain operating conditions due to the heat transferred through the flame tube.

At high exhaust temperatures, the thermocouples start glowing. Hence, the exhaust temperatures had been corrected for the radiative heat loss using a model proposed by Príkopský [25]. The adiabatic exhaust temperature shown in Fig. 5 is calculated from the chemical equilibrium of the global mixture using GRI-MECH 3.0.

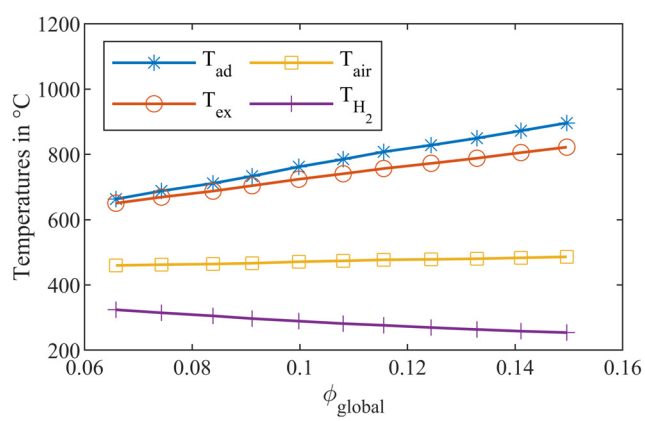

Fig. 5 Adiabatic, exhaust, air plenum, and H_2 injection temperature of a measurement series

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Table 3 Operating conditions of the laboratory tests

Property	Symbol	Value	Unit
Thermal power (LHV H_2)	P_{th}	24–54	kW
H_2 mass flow	$\dot{m}_{H_2}$	0.72–1.62	kg/h
Air mass flow	$\dot{m}_{air}$	368	kg/h
Equivalence ratio	ϕ_{global}	0.06–0.15	
H_2 temperature	$T_{H_2,in}$	20	°C
Plenum air temperature	T_{air}	450–520	°C

4 Air Distribution

The flow in the burner and the combustion chamber has been analyzed within phase 1 of the project. Please refer to Tanneberger et al. [24] for a detailed description of the flow field and its dependence on the geometrical parameters. Figure 2 depicts the absolute velocity of a sample flow field in the combustion chamber. From the left, the swirled fuel—air mixture enters the combustion chamber. The flow undergoes vortex breakdown forming an opening jet and a central recirculation zone. The premixed flame is supposed to stabilized in the strong shear layers between the recirculation zone and the high velocity jet (50–80 m/s). The secondary air injection consists of an axial cooling jet at the flame tube wall and the flow through the dilution holes, which can be seen as high velocity streaks in Fig. 3.

The splitting ratio of the air flow into primary and secondary air is determined by the effective cross sections of different flow paths inside the combustor, i.e., burner, dilution holes, and cooling slits. These cross sections are variable due to the setting of the geometrical parameters A and S (L has only minor effect) and due to the design of the flame tube, specifically the pattern and size of its dilution holes. To evaluate the flow distribution, the pressure of the cold flow is measured at the positions p_0 to p_3 in Fig. 6 with blockages of certain flow paths. From the measured pressure losses of the different paths, a model is set up to calculate the flow distribution for each parameter set and each flame tube. Please note that the model simplifies the real geometry and neglects the effect of the temperature rise due to the heat release in the flame. Hence, it resembles only an estimation of the air distribution. An example distribution is shown in Fig. 6.

In the example, 21% of the air is flowing through the mixing tube and the reaction zone of the flame as primary air. The remaining 79% is secondary air and does not take part in the reaction. This estimation can be used to calculate the local equivalence ratio ϕ_{local}, which only takes into account the primary air flow, as opposed to ϕ_{global} that takes the total air mass flow. In phase 1, ϕ_{P1} is not differentiated between local and global as there is no secondary air and therefore both are equal.

The first design of the flame tube (V1) is based on the original flame tube of the mGT and has six lines of dilution holes at different axial positions. In order to vary the ratio of primary and secondary air, six different flame tubes have been tested. For V2 to V5 some lines of holes have been closed to reduce the secondary air. The selection was done using computational fluid dynamics simulations. Flame tube V5* features the same open lines as V5 but the double line of small holes is further upstream. V5** is the same flame tube as V5* but the cooling slits in the front plate have been closed resulting in $\dot{m}_K = 0\%$ (Fig. 6). Table 4 lists the flow distribution for each flame tube along with the resulting local equivalence ratio for a fixed global one.

Table 4 Flow rates through the mixing tube (*M*), the cooling slit (*K*), and the dilution holes (*L*) for different flame tubes

Flame tube	M (%)	K (%)	L (%)	ϕ_{global}	ϕ_{local}
V1	17.1	13.3	69.7	0.12	0.68
V2	19.5	15.1	65.4	0.12	0.60
V3	19.0	14.7	66.3	0.12	0.62
V4	20.9	16.2	62.9	0.12	0.56
V5	21.8	16.9	61.3	0.12	0.54
V5*	21.6	16.8	61.7	0.12	0.54
V5**	24.8	0	75.2	0.12	0.47

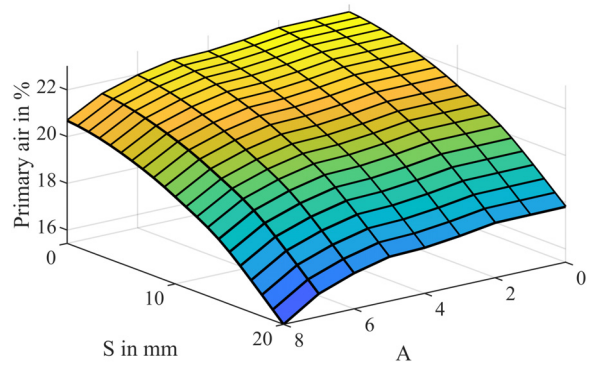

Fig. 7 Dependence of the primary air on the swirler blockage *S* and the axial injection *A* with flame tube V4

The influence of the geometrical parameters is as follows: The swirler blockage S, which is used to increase the swirl intensity, raises the pressure loss and therefore reduces the primary air mass flow as shown in Fig. 7. The same applies for the reduction of the axial air injection A, even though the effect is small up to $A = 5$.

The flow distribution model will be used in Secs. 6 and 7 to estimate the flow through the mixing tube and, hence, the local equivalence ratio. However, the flow coefficients have been recorded at cold flow conditions and without a flame. Therefore, the local equivalence ratio must be seen as estimation.

A direct influence of the secondary air injection can be seen in the temperature profiles of the exhaust gas shown in Fig. 8. The profile of phase 1 without secondary air shows the most homogeneous temperature distribution. In phase 2, the profiles are much more uneven, with the exception of flame tube V5, which has the lowest secondary air flow and therefore the lowest flame temperature. It is also the only flame tube where two rows of downstream holes are closed, which means that more of the secondary air is injected

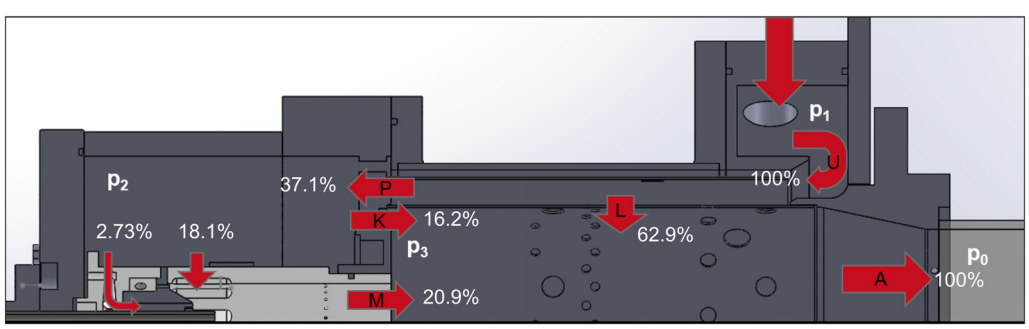

Fig. 6 Estimated air distribution in the combustor with flame tube V4

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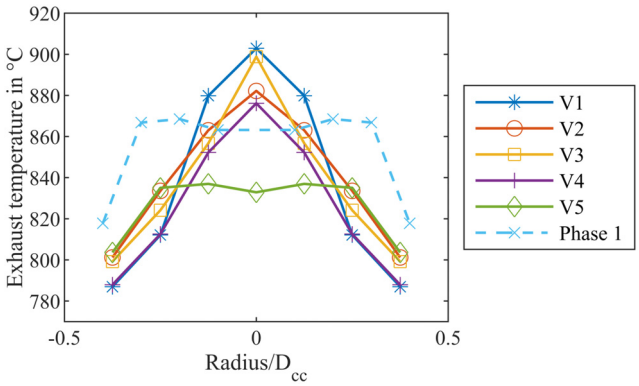

Fig. 8 Exhaust temperature profiles of phase 1 (ϕ_{P1}=0.2) and phase 2 (ϕ_{global}=0.15)

through the upstream holes, giving the flow more time to homogenize the temperature field. On the other hand, V3 shows the narrowest temperature profile. Here, one of the upstream rows is closed, which means that more secondary air is injected through the downstream holes, which are closer to the measuring position. Regarding the temperature profiles, V5 seems to be best suited for integration in the mGT as a uniform temperature distribution is generally desirable for the turbine inlet. However, the mGT in the project has a radial turbine, which is less sensitive than axial turbines regarding the temperature profiles.

5 Flame Shape and Stability Range

Due to the mGT like setup, there is no direct optical access to the flame in phase 2 of the project. Therefore, a library of flame shapes has been recorded in phase 1 using a fused silica combustion chamber and an OH* filtered camera (see Ref. [24]). Please note that no secondary air (neither dilution nor cooling) was applied in phase 1. However, since the flames are rather compact, the influence on the flame shape should be small. The previous publication [24] presented a parameter study of the flame shape regarding the equivalence ratio, the swirler blockage, the axial injection, center body position, and the share of central fuel injection.

The technically premixed H_2/air mixture generally burns in a V- or M-shape depending on the operating conditions and parameter settings. Figure 9 shows the shape of two stable flames for different equivalence ratios. However, geometry parameter changes can alter the flow field as described in Ref. [24], which may trigger flame flashback. Then, the flame propagates quickly upstream into the mixing tube. The flame stabilizes itself in the wake of the center body and keeps burning as a jet flame in the mixing tube and the combustion chamber. In the event of a flashback under laboratory conditions, the burner was not damaged and the flame could be restored by switching to the pilot flame [24].

The flame shapes of the other two fuel injection modes can be seen in Fig. 10. On the left, 50% of the fuel is injected via the central fuel

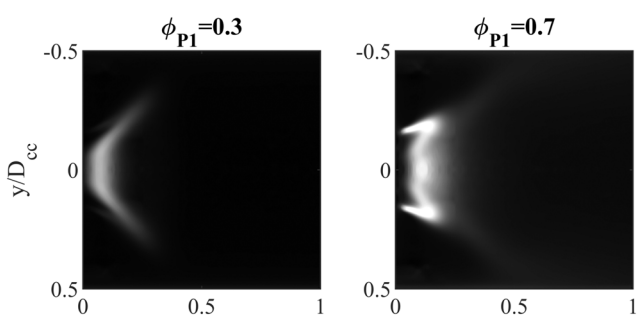

Fig. 9 Deconvoluted OH* intensity of the premixed hydrogen flame in phase 1 at two equivalence ratios

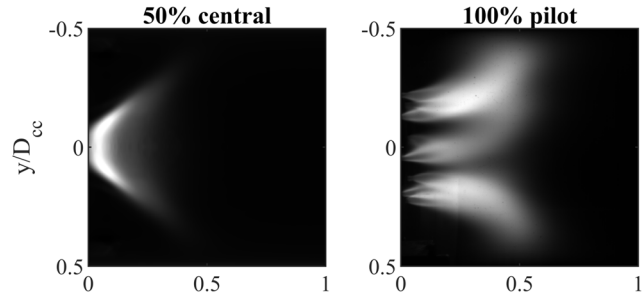

Fig. 10 Left: deconvoluted OH* intensity of a 50/50% premixed/central hydrogen flame in phase 1. Right: line-of-sight OH* intensity of the pilot hydrogen flame in phase 1.

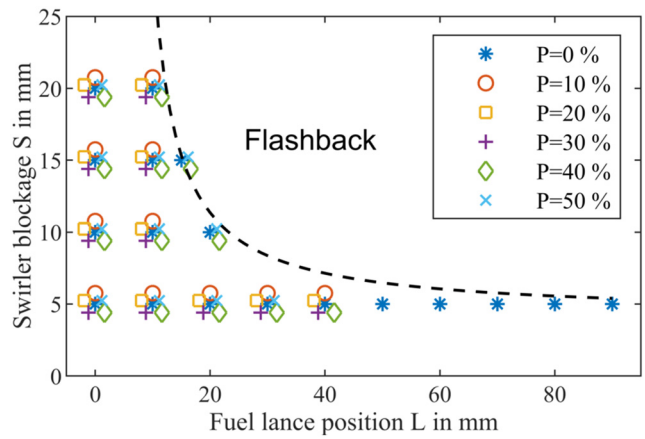

Fig. 11 Phase 1: stable operating points for variations of the parameters *S*, *L*, and *P* (central injector) at ϕ_{P1}=0.4 and *A*=0

lance, which increases the flame intensity in the center. The right image shows the pilot mode, which consists of eight individual jet flames.

Figure 11 depicts the operational range of the flame depending on three of the parameters. The different markers symbolize increasing levels of fuel share via the fuel lance (P). Hence, points with all five markers feature stability for a variety of fuel distributions and are very robust against flashback. The main driver of instability is the positioning of the fuel lance (L). For $L <= 10$ mm all flames are stable, while for higher L the stability declines with increasing swirl intensity as shown by the dashed line in Fig. 11. Due to flow field changes, L varies the fuel distribution and thus the local equivalence ratio and flame speed. However, these flow-induced changes in the concentration field are smaller than for a variation of P. Since the receptivity of flashback is still higher for L than for P, it can be concluded that the flashback is dominantly triggered by hydrodynamic effects. A more detailed description of Fig. 11 including the behavior close to the stability line can be found in Ref. [24].

The utilization of the burner in the mGT does require not only static but also dynamic stability. During the experiments in phase 1, the sound pressure levels had been monitored by a microphone in the exhaust tube. There are no indications of thermoacoustic instabilities under normal conditions at any of the tested parameter settings.

6 Emissions Hydrogen

This section analyses the NO_x emissions of the burner in phases 1 and 2 with pure H_2 fuel. The emissions are normalized to dry conditions and 15% O_2 in the exhaust using a normalization method proposed by Garan et al. [26], which takes the whole exhaust gas composition into account to avoid the artificial fuel-specific NO_x

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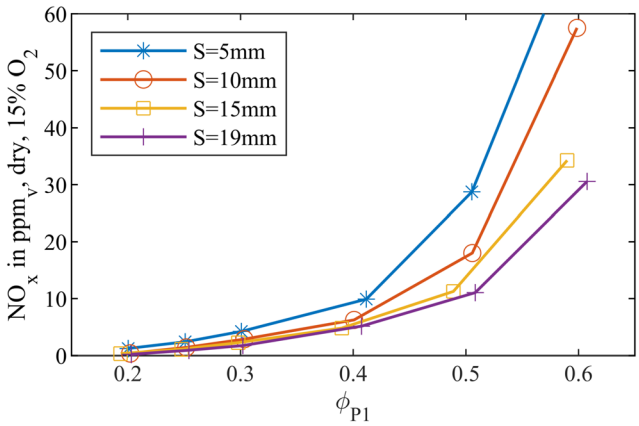

Fig. 12 Phase 1: NO_x emissions for different swirl intensities and A=0, L=10 mm, and P=0% [24]

increase of H_2 [27]. This method was already used in the previous publication [24].

$$c_a = \frac{1}{1 - X_{H_2O}} \frac{0.21 - X_{O_{2,ref}}}{0.21 - X_{O_2}} \frac{\alpha + 1 - 0.21}{\alpha_{ref} + 1 - 0.21} \frac{\beta}{\beta_{ref}} \frac{LHV_{ref}}{LHV} \tag{1}$$

The NO_x emissions of the flame without secondary injection have been measured and analyzed regarding the parameter influence in phase 1. The results are summed up in this paragraph. Please refer to Ref. [24] for more information. The strongest impact on the NO_x emissions has naturally the variation of the equivalence ratio as depicted in Fig. 12. Also significant is the influence of the swirl parameter S, which reveals a monotonic behavior as increased swirl intensity reduces the NO_x emissions significantly. Shifting fuel from premixed to central injection increases NO_x quickly first and then saturates at approximately $P = 50\%$. Shifting it to the pilot leads to a slower and more steady increase (Fig. 13). The parameter A and L have only little influence on NO_x when varied individually.

In phase 2, the burner features secondary air in the form of cooling and dilution injection as described in Sec. 3. The pattern of dilution holes in the flame tube determines the air distribution (see Table 4) and, hence, the local equivalence ratio of the flame. Figure 14 shows the influence of five different flame tubes on the NO_x emissions over the global equivalence ratio. All the emissions increase as expected with ϕ_{global}. However, the gradient depends strongly on the flame tube version, i.e., the dilution hole pattern. For high fuel flow the NO_x level of V1 is three times higher than for V5. The change can be mostly attributed to the declining primary air flow. Figure 15 shows that the lines for different flame tubes nicely collapse when plotted

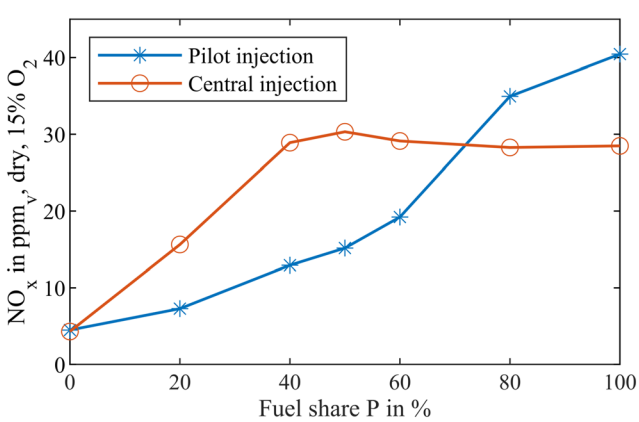

Fig. 13 Phase 1: evolution of the NO_x emissions when switching from premixed to pilot or central (L=10 mm) injection at ϕ_{P1}=0.4 [24]

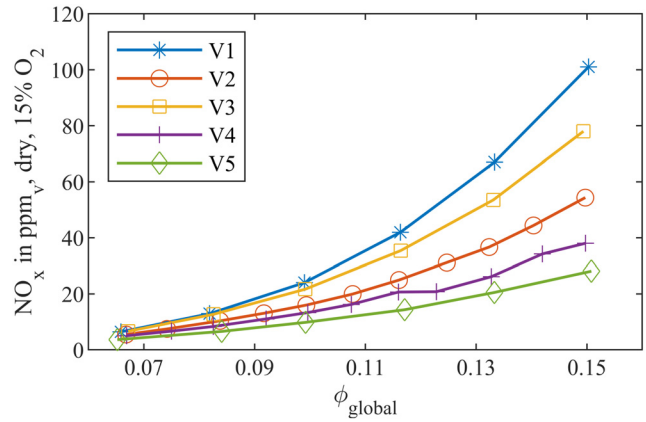

Fig. 14 Phase 2: NO_x emissions for different flame tubes over global equivalence ratio

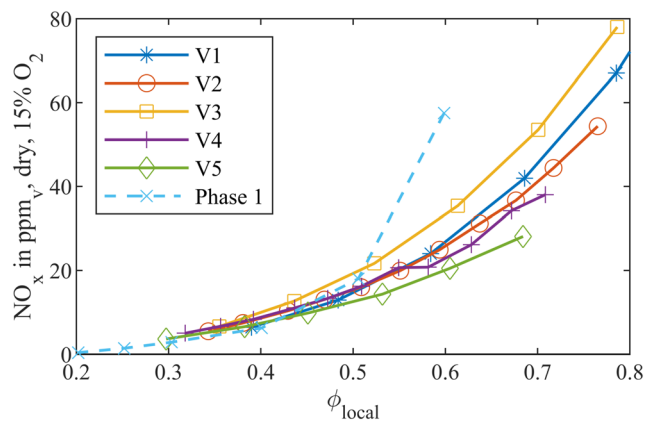

Fig. 15 Phase 2: NO_x emissions for different flame tubes over local equivalence ratio (-) and the NO_x emissions from the burner in phase 1 (- -)

over the local equivalence ratio (calculated from the estimated flow distribution). The collapsed emission lines also align with the equivalent burner configuration in phase 1, except for the last point. This indicates that the dilution of the exhaust flow plays a greater role at large ϕ_{local}, where the adiabatic flame temperature is very high and secondary air injection leads to rapid cooling and quenching of NO_x generation. The level of dilution does not appear to have any influence, as V1, V2, and V4 have the same NO_x emission over ϕ_{local}. The location of injection, on the other side, has a (small) effect. The flame tube with the highest emissions over ϕ_{local} is V3, the only flame tube in which one of the three upstream rows of holes is closed. This means that, compared to the other flame tubes, a higher proportion of the secondary air is injected further downstream. V2, for example, has a similar primary air flow as V3, but lower emissions, as more secondary air is blown in closer to the flame, which leads to faster cooling of the hot exhaust gases. This effect is more pronounced with larger ϕ_{local} due to the higher adiabatic flame temperature.

The flame tubes V1–V5 differ in the total size of the dilution hole area. V5 and V5* have the same hole lines, but at different locations. The third line (small double holes in Fig. 6) has been moved upstream to approximately 25 mm beyond the front plate. V5** has the same flame tube as V5*, but the cooling air slit has been closed ($\dot{m}_K = 0$), leading to a higher flow rate through the mixing tube (see Table 4). However, when plotted against the local equivalence ratio, all three configurations show no significant difference in NO_x emissions. This means, while the distribution of secondary air between the two clusters of dilution hole lines plays a (small) role,

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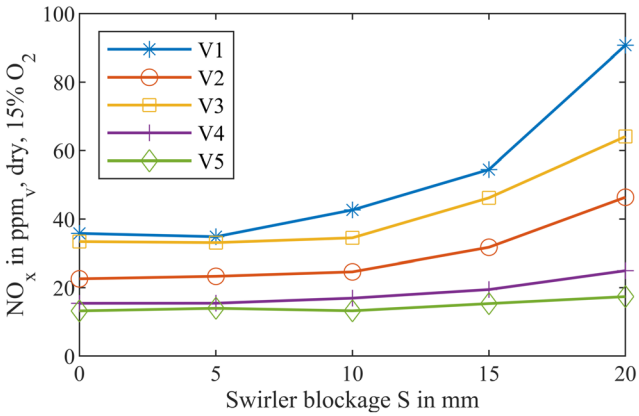

Fig. 16 Phase 2: NO_x emissions for different flame tubes over the swirler blockage *S*

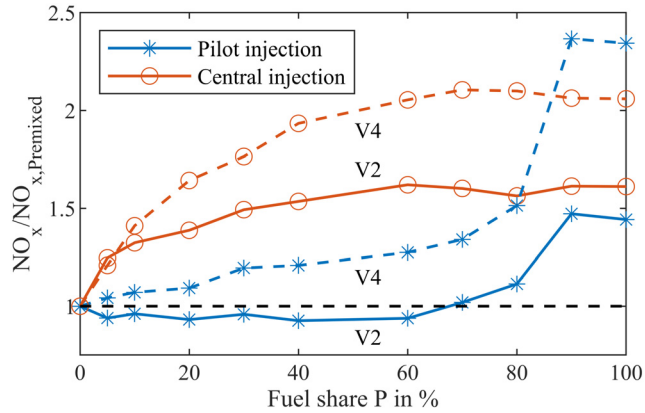

Fig. 17 Phase 2: evolution of the NO_x emissions when switching from premixed to pilot or central (L=10 mm) injection for V2 (-) and V4 (- -) at ϕ_{global}=0.115 [24]

the exact position of the holes within the first cluster is not significant.

Besides the flame tube design, the burner itself features geometrical parameters. The swirler blockage shows an interesting behavior compared to phase 1. All points in Fig. 16 were measured at constant thermal power and constant global equivalence ratio. An increase in emissions can be seen at higher swirl, which clearly contradicts the results from phase 1 (Fig. 12). In phase 1 the NO_x emissions show a reduction from $S = 5$ mm to $S = 19$ mm, which goes along with an increase in the pressure loss of 4 kPa ($\phi_{P1} = 0.4$) due to the larger swirler blockage. For the same parameter change in phase 2, the pressure loss only increases by 0.9 kPa (V5) because the blockage restricts only the flow through the mixing tube and the air can divert to the secondary air holes. Thus, the further blockage of the swirler leads to a decline in primary air as shown in Fig. 7. This increases the local equivalence ratio and, hence, the NO_x emissions. For example, with flame tube V5, ϕ_{local} rises from 0.51 at $S = 5$ mm to 0.63 at $S = 20$ mm with nearly constant emissions. Without the positive effect of the swirl intensity, such a step in the equivalence ratio would correspond to a significant increase in NO_x (Fig. 15). This effect is stronger the larger the dilution holes in the flame tube, because the ϕ variation takes place in a higher range of the local equivalence ratio with a larger gradient of emissions. A similar effect can be seen with the blockage of the axial air injection (parameter A). Please note that the effect does not occur in phase 1 because there is no secondary air and all the air flows through the mixing tube even when the pressure loss increases. This explains the opposite trend in Fig. 12.

Shifting the fuel from the premixed nozzles to pilot and central injection in Fig. 17 shows a similar trend as in phase 1 (Fig. 13). For the central injection the NO_x emissions increase quickly due to the richer center of the flow. The growth saturates at approximately $P = 50\%$. For the pilot injection, NO_x rises slowly first, followed by a sharp step up close to $P = 100\%$. The step appears approximately at the point, when the air/fuel mixture in the mixing tube reaches the blow off limit and would not be able to burn on its own anymore. The remaining hydrogen must be consumed by the pilot flames.

In terms of NO_x, the central injection is always worse than premixed, while the relation between pilot and premixed depends strongly on the primary/secondary air splitting because the pilot flame is not influenced in the same way by the primary air as the premixed flame. The 100% pilot flame can have even lower emissions than the premixed flame for high local equivalence ratios. The reason for the relatively low pilot emissions is to be seen in the miniaturization into eight single flames with high momentum injection in the pilot mode (see, e.g., micromixing concept [28]).

The operating conditions offer a second set of parameters besides the geometrical variations. The relevant parameters for the gas turbine are the operating pressure, the thermal power and the preheating temperature. While the pressure is fixed in the atmospheric pressure test rig, the thermal power can be varied in two ways. First, the fuel mass flow is increased at constant air mass flow resulting in a rise of the thermal power level and the global/local equivalence ratio. This type of variation is shown in Figs. 14 and 15 and displays increasing NO_x emissions. The second way is to hold the equivalence ratio constant and increase the air and fuel mass flow at the same time. Figure 18 shows that this way the emissions drop with rising thermal power. The oxygen concentration measured in the exhaust indicates that the equivalence ratio was not perfectly constant during the measurement series. However, this can be corrected (dashed line) and has only a small share in the observed decline. One reason for the decreasing NO_x could be that the higher flow velocity improves the premixing of air and hydrogen leading to a more uniform mixture profile in the flame which would reduce NO_x. The Reynolds numbers in the mixing tube increase from 35,000 to 76,000 for the 500 °C series. However, the same declining NO_x trend also appears for the pilot and central flame, which makes it unlikely that it is solely a mixing phenomenon. Another factor for the NO_x generation is the residence time in the hot zone. The large amount of dilution air in this combustor configuration (78%) confines the hot zone spatially. The higher transport velocity at higher power reduces the residence time in the hot zone and possibly leads to earlier quenching of the NO_x generation.

Figure 18 also shows the influence of the preheating temperature, which is increased in steps of 150 °C. The exhaust temperature of each line is fairly constant at roughly 450 °C, 550 °C, 650 °C, and 750 °C. As expected, NO_x rises with the fresh gas temperature due to

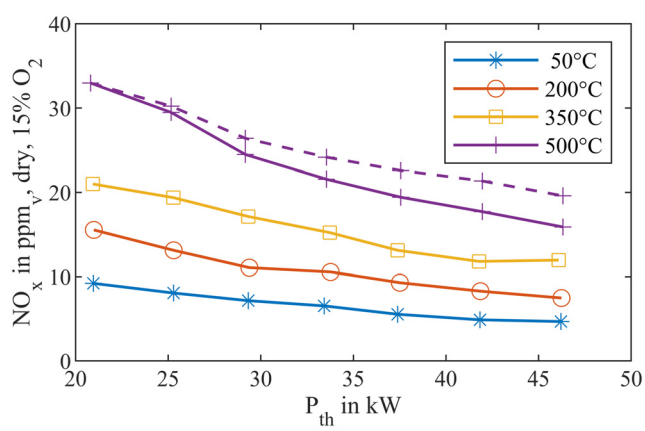

Fig. 18 Phase 2: variation of the thermal power (via H_2 mass flow) and preheating temperature (different lines) at constant ϕ_{global}=0.115 with flame tube V5*

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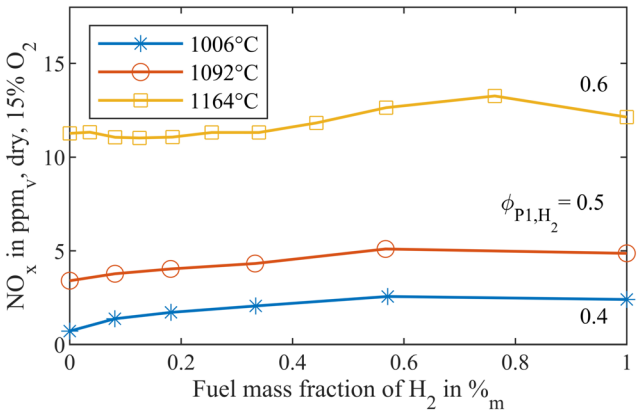

Fig. 19 Phase 1: NO_x of H_2/NG mixtures at three different exhaust temperatures

thermal effects in the NO_x formation. However, the decline with increasing power can be found for each temperature line.

7 Emissions Fuel Mixtures

The scope of the project is to fuel the entire run of the mGT with H_2 in phase 3. However from a scientific and technological perspective, it might be also interesting to test mixtures of H_2 and NG in the mGT. Therefore in phase 2, the burner is tested in the test rig with natural gas as well. H_2 and NG are mixed in the supply line to the burner with respect to a constant thermal power level and constant exhaust temperature. These tests have been done in phase 1 and phase 2 with different results.

The findings of phase 1 regarding fuel mixtures have been reported in Ref. [24]. Figure 19 shows the evolution of NO_x emissions from 100% NG to 100% H_2 for a constant burner configuration. When normalized using the fuel-corrected method (Eq. (1)), the NO_x emissions show no significant incline from NG to H_2. The three points with 100% H_2 correspond to $\phi_{P1,H_2} = 0.4$, 0.5, and 0.6, the three 100% natural gas points to $\phi_{P1,NG} = 0.52$, 0.64, and 0.79.

A similar series was recorded in phase 2 with flame tube V5* and the same burner settings as in phase 1. It is shown in Fig. 20. Please note that the exhaust temperatures are lower due to the secondary air injection in phase 2. Nevertheless, the local equivalence ratios are higher with $\phi_{local,H_2} = 0.53$, 0.61, and 0.76 and $\phi_{local,NG} = 0.71$, 0.77, and 0.96, respectively. In contrast to phase 1, the emissions are now rising. Even though all three curves indicate higher NO_x for H_2 compared to NG, they show slightly different behaviors. For the coldest flame, the emissions decline first before rising. The middle

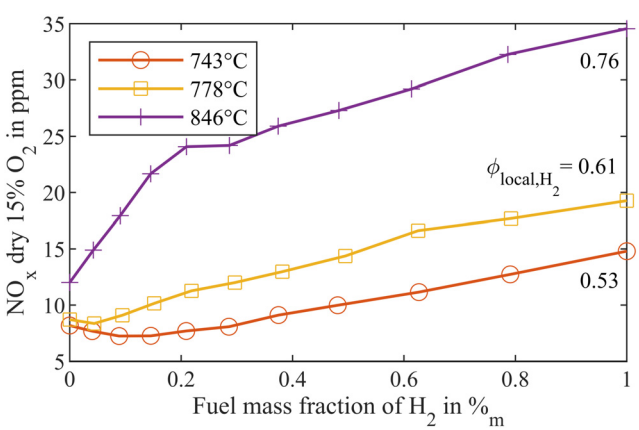

Fig. 20 Phase 2: NO_x of H_2/NG mixtures at three different exhaust temperatures

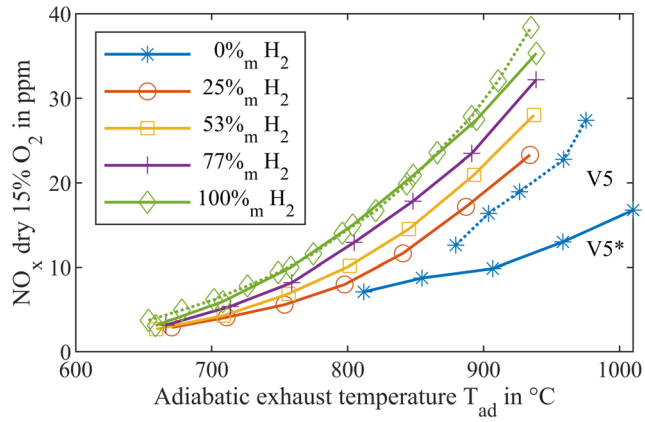

Fig. 21 Phase 2: NO_x of H_2/NG mixtures at different H_2 mass fractions. V5 (· ·) and V5* (-).

one increases linearly and the hottest flame picks up strongly up to $20\%_m$ and then similarly to the others.

The three measurement series lie in different ranges of equivalence ratios, which might be the reason for the different behavior. For clarification, additional series with constant mixtures but varied equivalence ratios have be measured by increasing the fuel mass flow at constant air flow for each line. Since H_2 and NG have different stoichiometry, these lines are plotted in Fig. 21 over the adiabatic exhaust temperature instead of the equivalence ratio. However, each line still presents an increase in the local equivalence ratio. For H_2 and the H_2 enriched mixtures, Fig. 21 shows the expected nonlinear rise of NO_x. The pure NG curve, however, exhibits a linear increase, which cannot be expected, especially when reviewing literature data for similar burners [16,17]. One possible explanation is that the size of the natural gas flame, which is less compact than the enriched flames (see Ref. [24]), interacts with the dilution air leading to reduced NO_x at high power. This is also indicated by a comparison of flame tube V5 and V5* (earlier dilution holes), which show the same NO_x values with H_2 but lower NO_x for V5* with NG (Fig. 21). Please note that the NG flames could not be measured for lower T_{ad} than shown in the plot due to blow-off, while the H_2-enriched flames could be operated even well below $T_{ad} = 650\,^\circ$C, showing the large operating range of the H_2 flame.

To investigate the opposing H_2–NG behavior of phases 1 and 2, it was tried to reproduce the findings of phase 1 by testing a flame tube without dilution holes (expect one for the spark plug) and without cooling air in the phase 2 setup. The increase in NO_x for H_2 enrichment is still present for this configuration and has a similar gradient as the 743 °C line in Fig. 20, which means that the secondary air injection seems not to be responsible for the different behavior between phase 1 and phase 2. Since this configuration and the phase 1 measurements differ mainly in the combustion chamber wall material and its surrounding, the difference might be caused by heat transfer and radiation properties. However, this point needs further investigation.

Figure 20 shows an increase in NO_x over the whole range of H_2 mass fraction. In the literature, H_2 enrichment often refers to the H_2 share in volume fraction. If the same data are plotted over the volume fraction, the NO_x increase occurs mainly above $80\%_v$, which many studies do not reach. Hence, the mass fraction with a steadier gradient might be the better metric to investigate the influence of H_2 enrichment.

8 Machine Test Preparation

Besides the investigation of the NO_x emissions, the tests in phase 2 seek to prepare the H_2 combustion tests in the actual mGT. First of all, the ignitability of the burner needs to be proven. Figure 22 shows the ignition window regarding the air and H_2 mass flow for two air inlet temperatures. Successfully ignited pilot flames are indicated by

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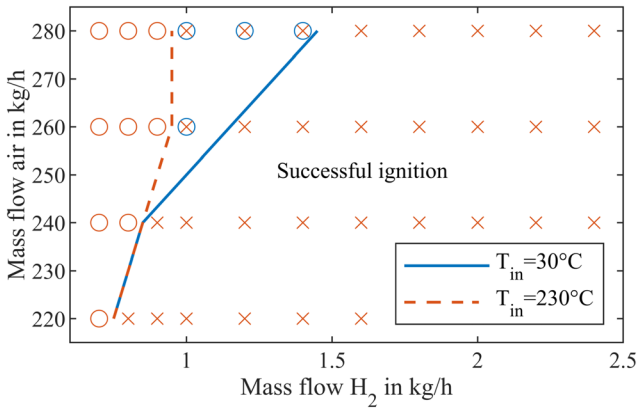

Fig. 22 Ignition tests in the atmospheric test rig for two air inlet temperatures: o no ignition, x successful ignition

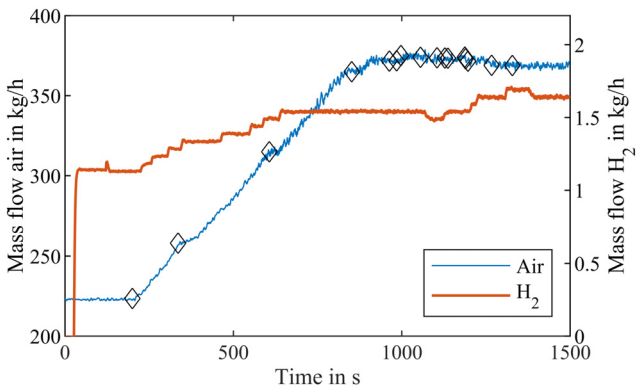

Fig. 23 Mass flows of the start-up test in the atmospheric test rig

an x, while points without ignition are marked with a circle. In general, the ignition window is sufficiently large for a cold start ($T_{in} = 30\,^\circ$ C) and even larger for hot starts after previous operation ($T_{in} = 230\,^\circ$ C).

After ignition, the mGT must run up to full load. To ensure a stable start-up, this process is simulated in the atmospheric test rig as shown in Figs. 23 and 24. The flame is ignited at $t = 25$ s in the 100% pilot mode. After 120 s, the fuel injection is switched from pilot to premixed. Subsequently, the air mass flow is increased to begin the start-up process. Since the operating pressure cannot be increased in the test rig, the air mass flow setpoints from the mGT are scaled to atmospheric conditions. The scaled setpoints are shown by the black diamonds in Fig. 23. At the same time, the air inlet temperature in the machine climbs due to higher compression and recuperation. It is

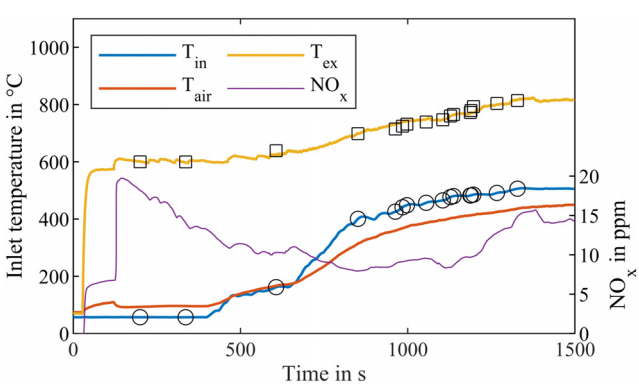

Fig. 24 Temperatures of the start-up test in the atmospheric test rig

shown in Fig. 24 by the black circles and recreated by increasing the preheater power in the test rig. The H_2 mass flow is adjusted to match the target turbine inlet temperature from the mGT (black squares).

The flame shows stable operation in the premixed mode from the fuel switch to full load. The necessary H_2 mass flow increases during start-up, but less strongly than the air mass flow. This results in a declining equivalence ratio, which can be seen in the declining NO_x time trace. The large peak of NO_x in the beginning can be avoided by operating the pilot flame longer before switching to premix mode. Please note that in the test bench, it is also possible to run the entire process in pilot mode. This may not be possible in the mGT, as the injector pressure loss for pilot is much higher than for premixed (see Table 2).

9 Conclusion

The paper presents the results of the second phase of the H2mGT project, which aims to develop and validate a hydrogen burner for a microgas turbine. In phase 2, the burner is investigated at atmospheric conditions in a machine like setup featuring the injection of dilution air. It is shown that the burner can be stably operated for multiple parameter settings with 100% hydrogen fueling. The influence of four geometric parameters as well as the flame tube's dilution hole pattern on the flame is presented. The size of the holes defines the ratio of primary and secondary air and, thus, the flames local equivalence ratio. This has a large influence on the NO_x emissions and the exhaust temperature profiles, both getting better with increasing primary air flow. The limiting factor here is the pressure loss of the burner, which also increases due to the smaller dilution holes. Reviewing the temperature profiles and emission data, flame tube V5 seems to be best suited for the integration into the mGT in phase 3. It features around one-third less dilution hole area than the original flame tube of the mGT, which is used to for the combustion of NG and diesel (with a different burner). Its pressure loss is about 1.3 kPa higher than flame tube V1.

An interesting aspect of the project is the comparison of phase 1 and phase 2: the fused silica combustion chamber setup (often used in combustion research) and the burner with secondary air injection. The NO_x analysis showed significant difference between both setups. First, the influence of the swirler blockage is reversed because it triggers a shift in flow distribution, which has a larger effect on NO_x than the swirl intensity. A second change can be found in the fuel transition from natural gas to hydrogen. While in phase 1, NO_x remained nearly constant, an increase was measured in phase 2. The magnitude of this increase depends on the local equivalence ratio and gets larger with higher fuel mass flow. In each case, the strongest gradient occurs above 80%$_v$ H_2 by volume.

At the end, the ignition and start-up process was successfully conducted at atmospheric conditions in preparation of the mGT tests, which will take place in phase 3 of the project.

Funding Data

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

Latin Letters

c_a = fuel-specific NO_x correction factor
D_{cc} = combustion chamber diameter (mm)
D_{mt} = mixing tube diameter (mm)
L_{cc} = combustion chamber length (mm)

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L_{mt} = mixing tube length (mm)
$\dot{m}_{air}$ = fresh air mass flow (kg/h)
$\dot{m}_{H_2}$ = hydrogen fuel mass flow (kg/h)
P_{th} = thermal power (kW)
X = mole fraction

Greek Symbols

ϕ_{global} = global equivalence ratio in phase 2
ϕ_{local} = local equivalence ratio in phase 2
ϕ_{P1} = equivalence ratio in phase 1

Abbreviations

LHV = lower heating value (MJ/m^3 or MJ/kg)
mGT = microgas turbine
NG = natural gas
NO_x = nitrogen oxides

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